

Bridging India's EV Charging Infrastructure Gap

An Investment-Priority Model Using SQL, Python, and Public/Official Datasets

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Focus: Energy Analytics & Decision Models

1. EXECUTIVE SUMMARY & STRATEGIC THESIS

India's electric vehicle (EV) transition is accelerating rapidly, but public charging infrastructure deployment remains highly fragmented across states. Capital allocation has not tracked market demand evenly, leaving high-growth adoption hubs with acute access deficits. This project evaluates state-level charging access across India using official parliamentary records, government vehicle registries, and crowdsourced geospatial data to construct a data-driven investment ranking for private and public infrastructure developers.

2. DATA SOURCES & ARCHITECTURE

- Vahan Dashboard (MoRTH):** Multi-year EV registration records (2019–2024) at office/RTO granularity (ground truth for vehicle demand).
- OpenChargeMap (OCM):** Crowdsourced geospatial charging station point API (used for localized spatial mapping).
- Ministry of Heavy Industries (Lok Sabha Data):** Official state-level charging station counts published via parliamentary response (ground truth for actual charging supply).

3. DATA QUALITY AUDIT & ANALYTICS ENGINEERING

- Crowdsourced Bias Mitigation:** Cross-referencing OCM against official government records revealed OCM captures only ~7% of actual stations nationally, heavily biased toward Southern states (0% to 64% coverage range). Official figures were isolated as primary ground truth, restricting OCM strictly to visual mapping.
- Distribution-Based Anomaly Removal:** Identified scraping errors where isolated local RTOs reported >200,000 monthly registrations. Applied a 10,000 monthly registration cutoff (derived from a 99.5th percentile distribution break) to eliminate false spikes—preventing a ~10x false inflation in Rajasthan's market size.
- Temporal Distortion Correction:** Restructured SQL window functions (using `LAG()`) to compute YoY momentum across complete calendar years (2022 → 2023), eliminating skew from partial-year 2024 data (Jan–May).
- Demand Alignment:** Excluded non-plug-in *Strong Hybrid EVs* to evaluate true public grid-charging infrastructure demand accurately.

4. METHODOLOGY: STATE PRIORITY INDEX

The composite Priority Score (normalized from 0 to 1) synthesizes three weighted signals into a unified index:

$$\text{Priority Score} = 0.5 \times \text{Gap}_{\text{norm}} + 0.3 \times \text{Scale}_{\text{norm}} + 0.2 \times \text{Growth}_{\text{norm}}$$

- Infrastructure Gap (50% Weight):** Inverted ratio of official charging stations per 1,000 registered plug-in EVs (higher deficit = higher priority score).
- Market Scale (30% Weight):** Total volume of active EV registrations (ensures capital targets viable markets).
- Adoption Momentum (20% Weight):** Full-year YoY adoption growth rate (2022 → 2023).

5. HEADLINE RECOMMENDATION: TOP 5 PRIORITY STATES

RANK	STATE	EVS REGISTERED (2024)	STATIONS / 1K EVS	YOY GROWTH	PRIORITY SCORE
#1	Uttar Pradesh	133,924	17.29	70.9%	0.819
#2	Chandigarh	2,476	5.25	135.6%	0.701
#3	Bihar	41,680	12.43	58.2%	0.622
#4	Rajasthan	78,204	19.37	31.0%	0.617
#5	Assam	25,879	13.33	41.7%	0.554

6. BUSINESS IMPACT & ACTIONABLE INSIGHTS

Key Strategic Takeaway: Uttar Pradesh (UP) emerges as India’s highest-priority investment target. UP leads the nation in total EV volume (>133k registrations) and maintains robust growth (>70% YoY), yet offers only ~17 stations per 1,000 EVs.

Directing capital expansion to UP addresses the largest absolute shortfall between EV adoption and public charger availability—mitigating range anxiety, unlocking fleet electrification potential, and maximizing ROI per deployed charger.